Microcontroller-Based Overvoltage Protection System

1. System Overview

This embedded design acts as an automated AC line voltage monitor and high-voltage cutout protection relay. It continuously senses single-phase mains voltage through a step-down signal conditioning stage, displays real-time RMS voltage and set points on a 16x2 character LCD, and drives an active-high relay to disconnect AC loads whenever mains voltage reaches or exceeds a user-adjustable trip threshold.

Circuit Subsystems & Working Principle

A. Sensing & Signal Conditioning

- **Galvanic Isolation & Reduction:** TR1 scales down high-voltage AC mains (230V RMS nominal) to a safe 12V AC RMS level.
- **Rectification & Smoothing:** Bridge rectifier D2–D5 converts AC to pulsating DC. Capacitors C1 and C2 filter high-frequency AC ripple into a steady DC voltage proportional to input AC amplitude
- **Attenuation & ADC Protection:** Potentiometer RV2 scales down the DC voltage into the 0–5V ADC range. Zener diode D6 clamps the A0 line to preventing transient overvoltage from damaging the microcontroller

B. Microcontroller Processing & Interface

- **ADC Conversion:** Analog Pin A0 measures the conditioned DC level. The software averages 30 raw samples to suppress residual power line ripple.
- **Calibration Mapping:** $V_{AC} = V_{ADC} \times 73.35$ (derived from $217VAC \approx 2.30VDC$)
- **User Input & Memory:** Debounced pushbuttons on Digital Pins 8 and 9 adjust the overvoltage cutoff limit in $\pm\ 5V$ steps. Adjusted limits save directly to EEPROM.

C. Protection Output Stage

- **Normal Mode ($V_{AC} < V_{Trip}$):** Pin 10 outputs HIGH (5V), holding Relay RL1 to power load L1.
- **Trip Mode ($V_{AC} \geq V_{Trip}$):** Pin 10 drops LOW (0V) instantly, Relay RL1 to isolate load L1.

System Calibration & Operation Procedure

1. **Power-On Inspection:** Ensure AC mains input is connected to TR1 primary terminals and verify 5V DC logic power supply at UNO1.
2. **Voltage Calibration:** Apply a known AC RMS voltage (e.g., 217V AC from source V1). Adjust trimmer potentiometer RV2 until the voltage displayed on Row 0 of LCD1 matches the reference meter reading.
3. **Threshold Setting:** Press INCREMENT (Pin 9) or DECREMENT (Pin 8) to set the trip threshold (e.g., 230V).
4. **Trip Verification:** Increase AC supply V1 above the set threshold. Confirm that Relay RL1 opens immediately, indicator light L1 turns off, and LCD1 reads Relay:OFF (TRIP)